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QuanSyn: A Package for Quantitative Syntax Analysis

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ABSTRACT

Over the past two decades, quantitative models and statistical methods have been increasingly applied to syntax research, with treebanks emerging as a vital resource for this field. However, tools specifically designed for calculating syntactic metrics remain relatively scarce compared to traditional word frequency measures. This gap hinders the deeper integration of quantitative methods into syntax research. In response, we have developed a tool called *QuanSyn*, which is tailored for quantitative syntax analysis. Based on treebanks, *QuanSyn* can calculate metrics such as dependency distance, hierarchical distance, dependency direction, and valency-related syntactic metrics. Additionally, it facilitates the construction of linguistic networks and rapid fitting of functions. *QuanSyn* can help lower the barriers to accessing syntactic quantitative analysis, thereby promoting the broader application of quantitative methods in syntax research.

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1. Introduction

Over the past two decades, quantitative models have been increasingly employed in syntax research, leading to the emergence of syntax measurement as a significant area within quantitative linguistics. The standard process of syntax measurement involves selecting a syntactic framework and then quantifying syntactic properties (Köhler, 2012). The advent of large-scale syntactically annotated corpora, such as treebanks in Universal Dependencies (UD),¹ has facilitated the rise of quantitative research on dependency and valency structures. Based on treebanks, many statistical patterns of human language have been discovered, such as *dependency distance minimization* (DDM) (Ferrer-i-Cancho, 2004; Futrell et al., 2015; Liu, 2008; Temperley, 2007) and *dependency direction continuum* (Liu, 2010; Tesnière, 1959). Furthermore, treebanks can be modelled as syntactic dependency networks, where network indicators can reflect syntactic patterns at the system level of language (Ferrer-i-Cancho et al., 2004).

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These treebank-based metrics are widely used in fields such as language classification (Liu, 2010) and cognitive computing (Liang et al., 2017), effectively reflecting the universality and particularity of languages. These studies highlight the importance of treebanks as vital resources for examining linguistic structural patterns and evolutionary principles. To be specific, treebanks serve as an interface between quantitative linguistics (Liu, 2009), computational linguistics (Vadas & Curran, 2011), and artificial intelligence (Socher et al., 2013).

However, the lack of computational tools to analyse large-scale treebanks poses challenges in uncovering hidden syntactic patterns. While quantitative linguistics boasts a variety of text analysis and corpus retrieval tools, such as *QuitaUp* (Cvrček et al., 2020), *AntConc* (Anthony, 2005), and *WordSmith* (Scott, 2008), few are specifically designed for syntax measurement. These tools primarily support traditional frequency-based metrics and are not well-suited to structured corpus resources.

A significant obstacle to the broader adoption of quantitative methods in syntax research is the limited availability of specialized tools. *Linear Arrangement Library* (LAL) has partially addressed this demand by employing advanced algorithms to effectively compute and simulate tree-based linear arrangement problems (Alemany-Puig et al., 2021). However, *LAL* offers a limited range of metrics and places less emphasis on quantifying fine-grained features, such as *mean dependency distance* (MDD) for specific parts of speech or dependency relations. Thus, further development of refined tools remains essential for advancing quantitative syntax analysis.

This paper introduces the development of a syntax measurement tool, *QuanSyn*, which facilitates the quantification of dependency and valency structures, the construction of linguistic networks, and the verification of linguistic patterns.

2. Installation and Workflow

It is encouraged to use *QuanSyn* as a Python package in a programming environment. A user-friendly interface named *QuanSyn Mini* is developed to facilitate users without a programming background. It provides the basic functionalities of *QuanSyn* and can be downloaded from the release section on GitHub.² For the sake of readability, the codes are not discussed here in detail and we refer to our README³ and documentation.⁴ The following sections will focus on the usage of *QuanSyn Mini* instead.

The workflow of *QuanSyn* is shown in Figure 1. Rectangles in grey are modules or functions provided by *QuanSyn*. These functions will be introduced in detail in the corresponding sections.

For the convenience of development and management, this project is hosted on GitHub.⁵

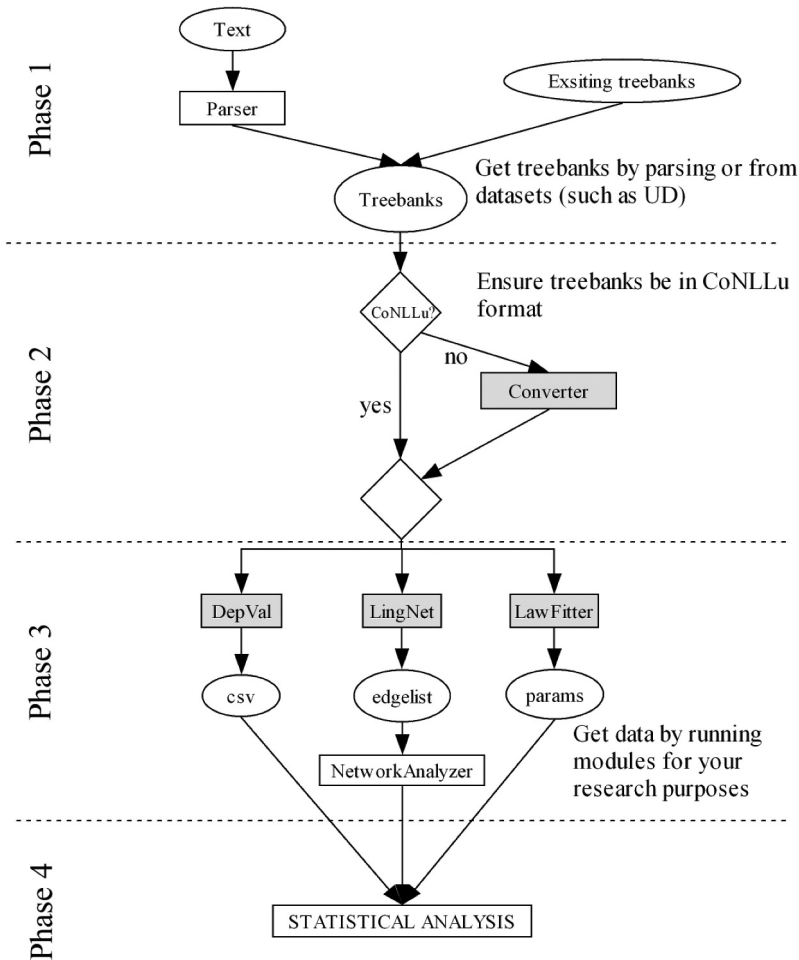


Figure 1. Flowchart of how to use *QuanSyn*.

3. Materials

QuanSyn supports only treebanks in CoNLL-U format.⁶ However, treebanks in other formats (e.g. CoNLL, PMT, MCDT) can also be analysed after format conversion by *Converter*, as shown in Figure 2.

Treebanks in CoNLL-U format can be obtained from datasets like UD or by performing dependency parsing using natural language processing tools such as *spaCy* (Honniбал & Montani, 2017) or *LTP* (Che et al., 2021). To illustrate this data structure, we will take a sentence derived from Parallel Universal Dependencies (PUD) as an example. To better align with linguistic syntactic traditions (Osborne & Gerdes, 2019), the corresponding treebanks are selected from Surface Syntactic Universal Dependencies (SUD). After

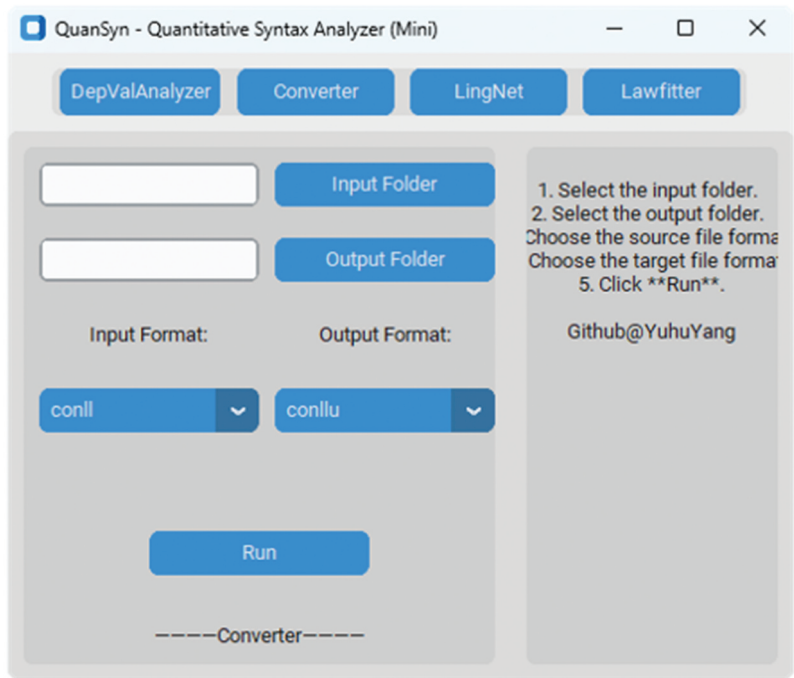


Figure 2. Converter and its usage in QuanSyn Mini.

Table 1. The CoNLL-U expression of the example sentence.

ID	FORM	LEMMA	UPOS	XPOS	FEAT	HEAD	DEPREL	DEPS	MISC
1	Maybe	maybe	ADV	RB	—	5	mod	—	—
2	the	the	DET	DT	—	4	det	—	—
3	dress	dress	NOUN	NN	—	4	compound	—	—
4	code	code	NOUN	NN	—	5	subj	—	—
5	was	be	AUX	VBD	—	0	root	—	—
6	too	too	ADV	RB	—	5	mod	—	—
7	stuffy	stuffy	ADJ	JJ	—	5	comp:pred	—	—

dependency parsing, the sentence *maybe the dress code was too stuffy* can be represented as shown in Table 1.

CoNLL-U format displayed in Table 1 is structured in columns, with each row representing a token and each column providing specific feature information for that token. For syntax measurement, the most basic features are *ID*, *FORM*, *UPOS*, *HEAD*, and *DEPREL*. These features can load basic information about dependency relations and valency structures of sentences. Based on them, a sentence can be transferred from a sequence to a tree. The *ID* column denotes the linear position of the word in the sentence. For example, *maybe* is the first word, so its *ID* is 1. *FORM* represents the word form, while *UPOS* indicates its part of speech. *HEAD* refers to the linear sequence number of the governor of

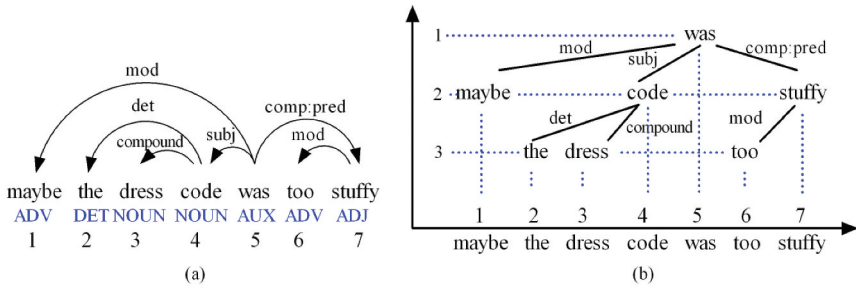


Figure 3. Two styles of dependency trees for the example sentence.

a given word. In this case, the governor of *maybe* is *was*, which has an *ID* of 5, so the *HEAD* for *maybe* is also 5. *DEPREL* specifies the type of dependency relations, such as the *mod* relation between *maybe* and *was*. The sentence in Table 1 can also be depicted as two styles of tree diagrams, as shown in Figure 3.

In Figure 3(a), the arcs represent dependency relations, with the arrows indicating the direction of these relations, pointing from governors to dependents. In Figure 3(b), the links between words are dependency relations. The vertical axis represents hierarchical levels of words, while the horizontal axis represents linear orders. Treebanks are databases that store such trees. The English treebank from PUD (PUD-E) contains 1,000 sentences, making it valuable for quantitative syntax research. We will use this treebank as an example to demonstrate the functions of *QuanSyn*.

4. Calculate Syntactic Metrics

Depval module consists of two main classes: *Converter* and *DepValAnalyzer*. *Converter* can help to transform treebanks into CoNLL-U format, which is friendly for syntactic metrics calculation by *DepValAnalyzer*. The interface and usage of *DepValAnalyzer* in *QuanSyn Mini* are illustrated in Figure 4.

Syntax measurement within the dependency grammar framework typically focuses on dependency relations and valency structures. Since some metrics assess features of words or dependency relations, others focus on sentence-level features, and yet others on text-level features, we have categorized the metrics into three hierarchical levels. Additionally, we include the distribution of metrics and *probabilistic valency patterns* (PVP), making a total of five dimensions. A detailed explanation is as follows.

4.1. Word/Dependency-Level Metrics

The word/dependency level includes four metrics, which calculate the syntactic properties of dependency relations or the words acting as dependents

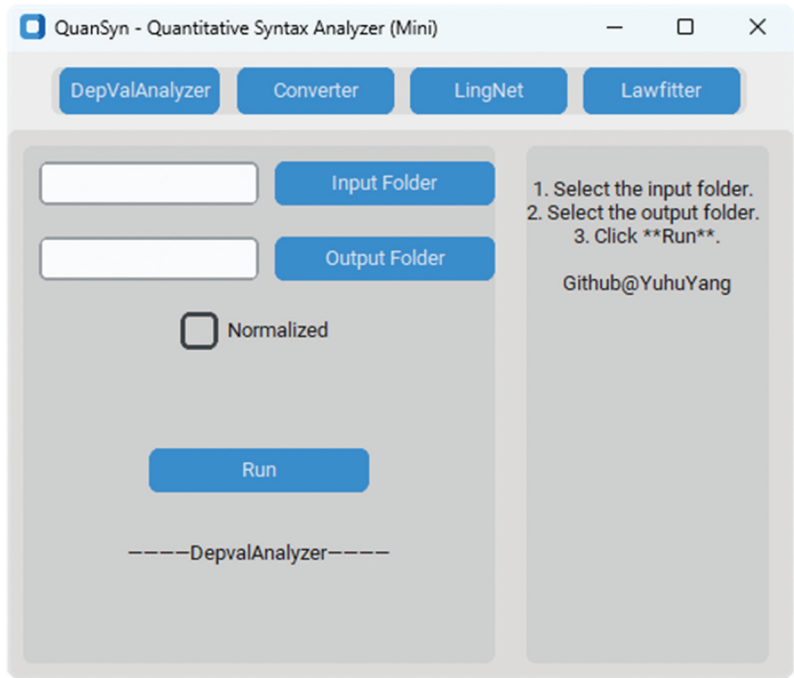


Figure 4. *DepvalAnalyzer* and its usage in *QuanSyn Mini*.

within these relations. Metrics and their definitions are listed in Table 2. To clarify these metrics, we calculated them using the example sentence in Figure 3.

When processing sentences, words must be processed sequentially, one at a time. A word remains in working memory until its governor has appeared, imposing a continuous memory load. The greater the distance between words and their governors, the heavier the cognitive load tends to be (Ferrer-i-Cancho, 2004; Hudson, 1995; Liu et al., 2017). These cognitive constraints can be reflected by both DD and HD, which serve as metrics of syntactic complexity. However, these two metrics have different focuses,

Table 2. Metrics in word/dependency-level.

Metrics	Definitions
<i>Dependency Distance (DD)</i>	The linear distance between the two words in a dependency relation. For instance, the DD of <i>maybe-was</i> is $ 1-5 = 4$.
<i>Hierarchical Distance (HD)</i>	The path length from a given word to the root in the syntactic tree. For instance, the HD of <i>maybe</i> is 1, and that of <i>the</i> is 2.
<i>Dependency Direction (DDir)</i>	The relative placement of a word to its governor. DDir is classified as <i>head-initial</i> (HI) if the word follows its head and <i>head-final</i> (HF) if the word precedes its head. For instance, the DDir of <i>maybe-was</i> is HF.
<i>Valency (V)</i>	The number of words that have a dependency relation with a given word. For instance, the V of <i>was</i> is 3.

reflecting two distinct aspects of language processing difficulty. DD focuses on comprehension, serving as a metric of the difficulty in transforming a linear sequence into a hierarchical structure (Liu, 2008). While HD is oriented towards production, measuring the difficulty of converting a hierarchical structure into a linear sequence (Jing & Liu, 2015; Liu, 2011).

Word order is a key typological feature of languages (Greenberg, 1963). In dependency grammar, DDir is closely associated with word order. Tesnière (1959) classified Indo-European languages based on dependency direction into *mitigated centrifugal* and *mitigated centripetal* types, roughly corresponding to HI and HF types. He assumes that Indo-European languages are distributed along a continuum formed by these two dependency directions. Based on a statistical analysis of cross-linguistic treebanks, Liu (2010) proposed the *dependency direction continuum*, which confirmed Tesnière's hypothesis. Dependency direction is thus regarded as a novel and effective typological metric.

Valency narrowly refers to the number of slots a verb opens in a specific syntactic context, which are filled by complements (Herbst, 2007). Broadly, this term refers to a word's potential capacity to combine with other words. In a sentence, this potential *static valency* is realized as explicit *dynamic valency*, with the valency count corresponding to the number of dependency relations a word participates in (Liu, 2011). It is also referred to as *generalized valency* (Liu, 2011) or *full valency* (Čech et al., 2010). Our understanding of *valency* leans towards the latter definition, treating it as a metric of a word's combinatory capacity.

4.2. Sentence-Level Metrics

Metrics at the word/dependency level can be averaged to generate sentence-level ones, which then reflect the syntactic features of entire sentences. In addition, some sentence-level metrics can be derived through summation or holistic computation. We have listed the sentence-level metrics in Table 3. The formulas for TW and TH were added by us, based on the descriptions in Zhang and Liu (2018).

MDD and MHD are sentence-level projections of DD and HD, respectively. The differences between them have been explained in Section 4.1. Due to general cognitive constraints, DD tends to be minimized (Ferrerri-Cancho, 2004; Futrell et al., 2015; Liu, 2008; Temperley, 2007). One significant manifestation of DDM is the limitation of MDD, with a threshold of 4 (Liu, 2008). It aligns with Cowan's (2001) idea that the capacity of short-term working memory is 4. This threshold also applies to MHD, indicating that both MDD and MHD possess psychological reality in the dimensions of language comprehension and production. In addition, moreover, linking syntactic distance with cognitive difficulty has been increasingly supported

Table 3. Metrics in sentence-level.

Metrics	Equations	Definitions
Mean Dependency Distance (MDD)	$MDD = \frac{1}{(n-1)} \sum_{i=1}^{n-1} DD_i $	Mean of DDs in a sentence. For instance, $(4 + 2 + 1 + 1 + 1 + 2)/(7 - 1) \approx 1.83$.
Mean Hierarchical Distance (MHD)	$MHD = \frac{1}{(n-1)} \sum_{i=1}^{n-1} HD_i $	Mean of HDs in a sentence. For instance, $(1 + 2 + 2 + 1 + 2 + 1)/(7 - 1) = 1.5$.
Dependency Direction (HI&HF)	$HI = \frac{N_H}{N} \cdot 100\%; HF = \frac{N_H}{N} \cdot 100\%$	Proportions of two types of dependency directions in a sentence. For instance, $HI = 1/6 = 16.7\%, HF = 5/6 = 83.3\%$.
Total Dependency Length (TDL)	$TDL = \sum_{i=1}^{n-1} DD_i $	Sum of DDs in a sentence. For instance, $4 + 2 + 1 + 1 + 1 + 2 = 11$.
Mean Hierarchical Dependency Distance (MHDD)	$MHDD = \frac{n-1}{MAXHL}$	The ratio of the sum of hierarchical differences (fixed at 1) between two words in a dependency relation to the maximum hierarchical level. For instance, $(7 - 1)/3 = 2$.
Variance of Dynamic Valency (V[k])	$V[k] = V^2 - \langle v \rangle^2 = \frac{\sum_{i=1}^n v_i^2}{n} - (2 - \frac{2}{n})^2$	Variance of valencies in a sentence. For instance, $(1 + 1 + 1 + 9 + 1 + 1 + 16)/7 - (2 - \frac{2}{7})^2 \approx 0.78$.
Tree Width (TW)	$TW = \max_{x \in HD} count(x, HD)$	Number of nodes at the level with the most nodes in a dependency tree. TW of the example sentence is 3.
Tree Height (TH)	$TW = \max(HD) + 1$	Maximum of levels in a dependency tree. TH of the example sentence is 3.
Sentence Length (SL)	$SL = n$	Number of words in a sentence.

n – number of words in a sentence, N – number of dependency relations in a sentence, MAXHL – maximum of hierarchical levels.

by evidence from psychological experiments and large-scale corpora (Demberg & Keller, 2008; Futrell et al., 2015; Liu et al., 2017). Therefore, MDD and MHD are efficient in reflecting syntactic complexity both psychologically and computationally (Niu & Liu, 2022).

Another variant of sentence-level DD replaces averaging with summation, referred to as TDL or *sum of dependency distance* (SOD). TDL conveys similar information to MDD, but it is more dependent on SL (Futrell et al., 2015).

Chen et al. (2021) have keenly observed a lack of strict symmetry in the calculations of MDD and MHD. DD represents the linear distance between two words in a dependency relation. However, HD does not measure their vertical distance. Instead, it represents the path distance between a word and the root word. The vertical distance between two words in a dependency relation is called *hierarchical dependency distance* (HDD). Since dependency relations cannot span multiple levels, HDD is fixed at 1, which is why it is not included in the word/dependency-level calculation. Strict symmetry requires that the sum of HDD should be divided by the vertical sentence length (number or maximum of hierarchical levels). The final metric is called MHDD, which also suggests DDM.

$V[k]$ represents the variance of valencies among words within a sentence, describing the degree of variation in the combinatory capacity of words (Lu et al., 2018). $V[k]$ is positively correlated with MDD, and its magnitude determines the minimum value of MDD (Ferrer-i-Cancho, 2013). The difference in $V[k]$ reflects the asymmetry of dependency relations, providing a perspective for exploring DDM starting from words' combinatory ability.

Metrics based on DD, HD, and V all originate from local features such as words or dependency relations. Then map to the sentence level through statistical operations such as averaging, summing, or calculating variance. In contrast, TW and TH must be calculated based on an entire syntactic tree, capturing the two-dimensional geometric properties that reflect the sentence's overall structure. Zhang and Liu (2018) demonstrated that both TW and TH increase non-linearly with SL, suggesting that DDM may be a result of a competitive and cooperative relation between these two metrics. TW and TH provide opportunities to study DDM and syntactic structures holistically at the sentence level, opening up avenues for further exploration and development.

Figure 5 shows the distribution of several metrics on SL, with the data derived from *DepValAnalyzer*.

4.3. Text-Level Metrics

Text-level metrics have two sources. Some metrics are derived by dividing word/dependency-level metrics by their frequencies, such as MDD,

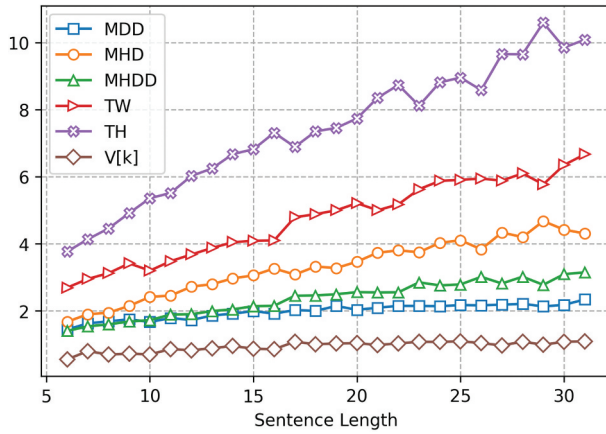


Figure 5. Metrics-SL distribution in PUD-E.

MHD, *mean valency* (MV), HI, and HF. Others are obtained by dividing sentence-level metrics by the number of sentences, including MHDD, *mean total dependency length* (MTDL), *mean sentence length* (MSL), $V[k]$, *mean tree width* (MTW), and *mean tree height* (MTH). MV is often used to evaluate the combinatory capacity of specific parts of speech rather than the valency of all words (Yan & Liu, 2022). A higher value indicates a stronger ability of that part of speech to combine with other words. The meanings of the other metrics are roughly consistent with the earlier descriptions and will not be introduced separately.

4.4. Distributions

Overall trends of metrics can be observed by analysing their probability distributions (DD, HD, V, SL, TW, TH) or rank-frequency distributions (dependency relations, parts of speech). Previous studies have shown that these distributions often follow some patterns (Liu, 2011; Petrini & Ferrer-I-Cancho, 2022; Yue & Liu, 2011).

Based on the data calculated by *DepValAnalyzer*, distributions are plotted in Figure 6.

4.5. Probabilistic Valency Pattern

The *probabilistic valency patterns theory* (PVPT) is a generalized valency theory that formalizes words or word classes in a feature-value structure (Liu & Feng, 2007; Yan & Liu, 2022; Yang & Liu, 2022). It is often illustrated as shown in Figure 7.

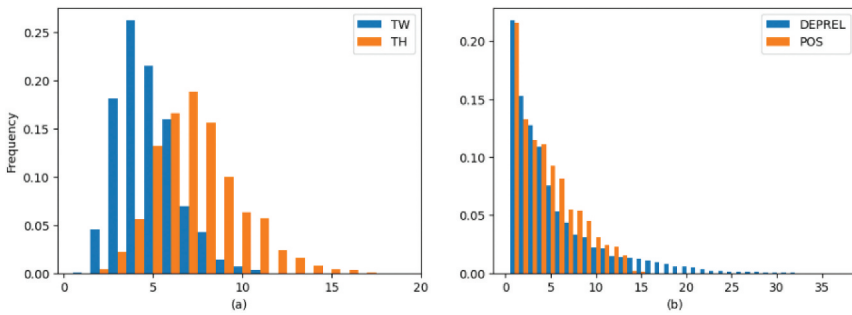


Figure 6. Distributions in PUD-E.

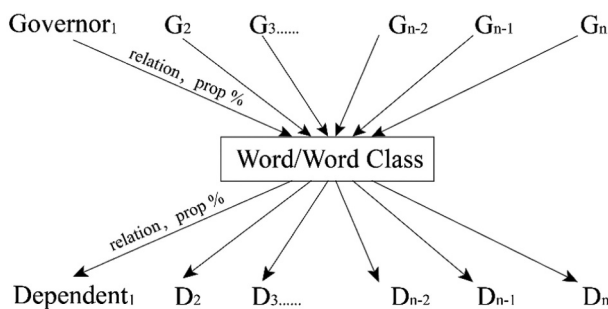


Figure 7. The PVP of a word or word class.

In Figure 7, rectangles represent words or word classes. Arrows denote labelled and directional dependency relations, pointing from governors to dependents. The labels indicate different types of dependency relations and their usage probabilities. Above the target word are all its governors in treebanks, while below are all its dependents.

For instance, *DepValAnalyzer* calculates the PVP of nouns. The top 5 high-frequency dependency relations and their corresponding probabilities are shown in Table 4.

5. Linguistic Networks

The view that language is a complex adaptive system has increasingly gained consensus (Five Graces Group et al., 2009). Complex networks have become a common approach to model language systems (Cong & Liu, 2014; Ferrer-i-Cancho & Solé, 2001). As mentioned in Section 4.2, TW and TH reflect sentence-level overall information. Similarly, language networks can provide insights into the overall structure and characteristics of a language. Linguistic networks typically treat words as nodes and relations among words as edges.

Table 4. A PVP sample of nouns of PUD-E.

act as a governor		act as a dependent	
dependencies	probability	dependencies	probability
det	29.8%	comp:obj	57.6%
mod	26.7%	subj	14.8%
udep	16.8%	compound	10.7%
compound	9.3%	conj	5.9%
mod@poss	4.4%	comp:pred	3.3%

Dependency relations and co-occurrence relations (i.e. the linear adjacency of words) are commonly used as the edges in networks. When dependency relations are treated as edges, a syntactic dependency network is constructed. While co-occurrence relations are used as edges, a co-occurrence network is obtained. The interface for extracting these two types of relations is shown in Figure 8.

Research has shown that linguistic networks, like most real-world networks, exhibit properties such as scale-freeness and small-world properties. Scale-freeness means that the degree distribution of nodes follows a power law, with the degree exponent typically ranging between 2 and 3 (Barabási & Albert, 1999). Small-world characteristics indicate the presence of central

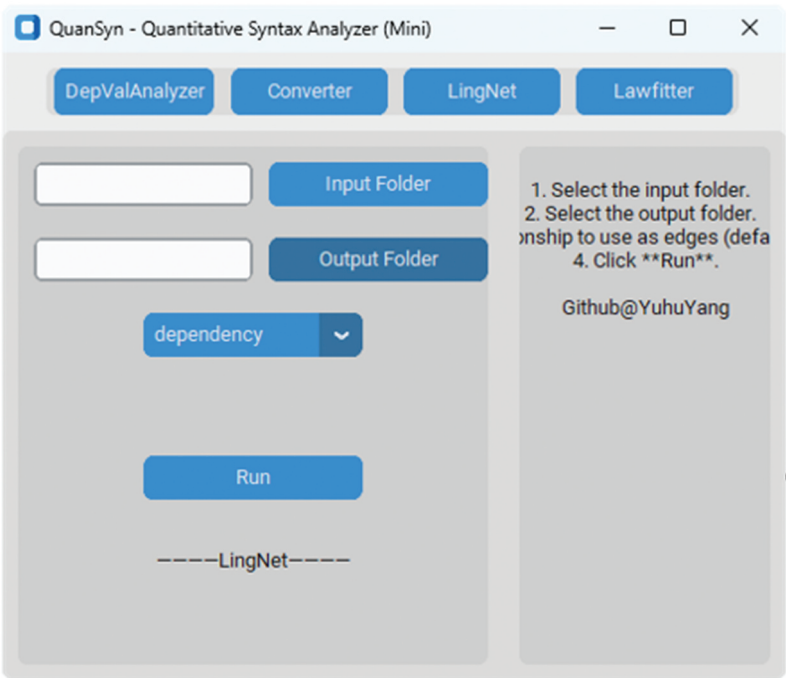


Figure 8. Lingnet and its usage in *QuanSyn Mini*.

Table 5. Network indicators based on PUD-E.

N	E	C	$\langle k \rangle$	γ	A	D
5720	16213	0.046	5.669	2.419	-0.119	0.001

nodes that significantly reduce the average distance between nodes in the network (Watts & Strogatz, 1998). Global indicators of linguistic networks have been applied in fields such as language classification (Abramov & Mehler, 2011), language acquisition (Monzalvo & Dehaene-Lambertz, 2013), and second language acquisition (Hao et al., 2021), demonstrating effective results.

Lingnet module can transform treebanks into edges based on dependency or co-occurrence relations and allow for the estimation of degree exponents. Once the network is constructed, network indicators can be calculated by tools such as *Networkx* (Hagberg et al., 2008). Part of the results based on PUD-E is presented in Table 5.

N - number of nodes, E - number of edges, C - clustering coefficient, $\langle k \rangle$ - average degree, γ - degree exponent, A - degree assortativity coefficient, D - density.

6. Fit Mathematical Models

The primary goal of extracting data from treebanks or networks is to uncover potential linguistic patterns. Quantitative linguistics has proposed and verified numerous laws, and their mathematical form can be used to fit language data. *Lawfitter* offers functions for several commonly observed language laws and also provides the flexibility for users to customize functions. The fitting functionality is implemented by *SciPy* (Virtanen et al., 2020). *Lawfitter* and its usage are illustrated in Figure 9.

For instance, we will attempt to fit the DD distribution in PUD-E using two different models, which are the power-law model and the linear model. Results can be seen in Figure 10.

The distribution of DD follows a significant Zipfian pattern, which has been widely validated (Liu & Feng, 2007; Petrini & Ferrer-I-Cancho, 2022). The R of the power-law model is significantly higher than that of the linear model, which means that our fittings are reliable. In addition, the negative R for the linear model is due to the data's inherent lack of a linear relationship.

7. Conclusion

QuanSyn is currently available as both a Python package and an application, making it accessible to users with and without a programming background. Of course, *QuanSyn* currently has some limitations, such as the lack of

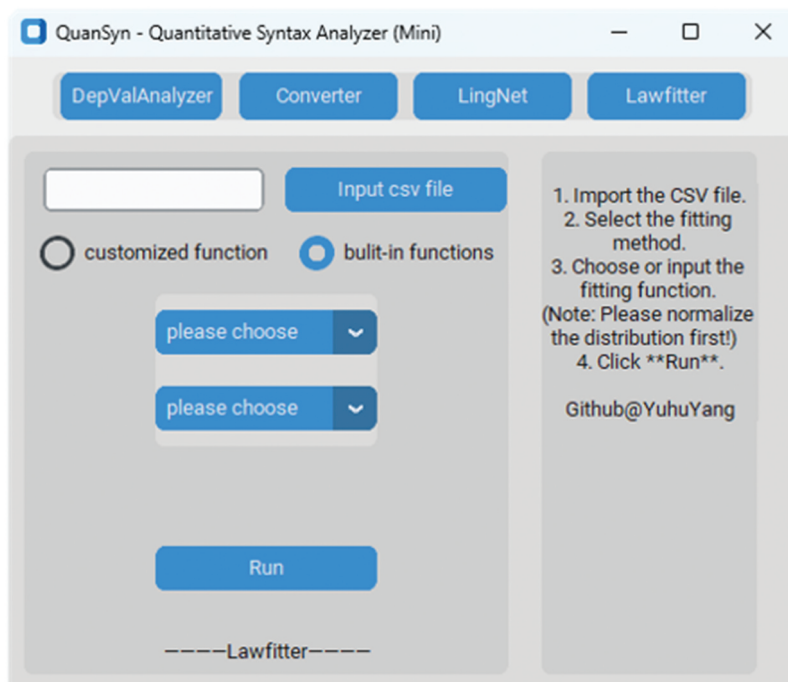


Figure 9. Lawfitter and its usage in *QuanSyn Mini*.

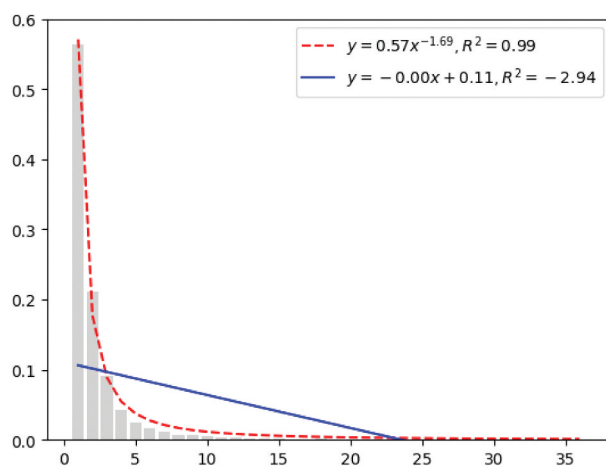


Figure 10. Fitting the DD distribution of PUD-E to two models.

visualization capabilities. In future development, we will continue to expand functionalities and optimize the algorithms. Quantitative linguistics seeks to explore structural patterns and evolutionary principles of language through precise mathematical models, with syntactic patterns being an important focus. *QuanSyn* can lower the barrier to quantitative syntax analysis, thereby facilitating research in this area.

Notes

1. <https://universaldependencies.org/>
2. <https://github.com/YuhuYang/QuanSyn/releases>
3. <https://github.com/YuhuYang/QuanSyn/blob/main/README.md>
4. <https://github.com/YuhuYang/QuanSyn/blob/main/document.md>
5. <https://github.com/YuhuYang/QuanSyn>
6. <https://universaldependencies.org/format.html>

Disclosure Statement

No potential conflict of interest was reported by the author(s).

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